LEARNING MUSIC STYLE FOR PIANO ARRANGEMENT THROUGH CROSS-MODAL BOOTSTRAPPING

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

What is music style? Though often described using text labels such as "swing," "classical," or "emotional," the real style remains implicit and hidden in concrete music examples. In this paper, we introduce a cross-modal framework that learns implicit music styles from raw audio and applies the styles to symbolic music generation. Inspired by BLIP-2, our model leverages a Querying Transformer (Q-Former) to extract style representations from a large, pre-trained audio language model (LM), and further applies them to condition a symbolic LM for generating piano arrangements. We adopt a two-stage training strategy: contrastive learning to align auditory style with symbolic expression, followed by generative modelling to perform music arrangement. Our model generates piano performances jointly conditioned on a lead sheet (content) and a reference audio example (style), enabling controllable and stylistically faithful arrangement. Experiments demonstrate the effectiveness of our approach in piano cover generation, style transfer, and audio-to-MIDI retrieval, achieving substantial improvements in style-aware alignment and music quality. [1]

1. INTRODUCTION

Automatic music generation is often controlled by *explicit* content such as melody, chords, and text labels [1–4], but music concepts can be more nuanced than we often realize. When musicians learn a style, instead of relying on abstract descriptors like "romantic" or "jazz" alone, they absorb patterns from music examples that share common stylistic traits. The commonality across these examples forms a style, an *implicit* one that cannot be fully described with words or labels but only understood through the music itself. This paper studies such implicit style qualities, particularly regarding *grooving patterns* and *dynamics* in piano performances, where abstract descriptors often oversimplify their richness, even though they are immediately perceivable in audio. We explore how such implicit style can be internalized from given audio examples and control music generation in a deep learning framework.

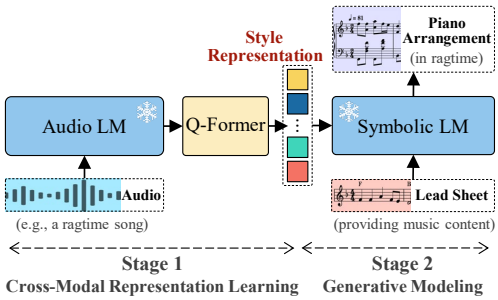


Figure 1: A Q-Former module bridges the modality gap between a frozen audio LM and a symbolic music LM. It extracts cross-modal music *style* from the hidden representations of the audio LM and, together with a lead sheet providing music *content*, conditions the symbolic LM for piano arrangement. The Q-Former is trained in a two-stage process, effectively bootstrapping audio-to-symbolic arrangement without re-training either LM backbone.

Large-scale music language models (music LMs) have shown strong capabilities in learning explicit music content, as demonstrated by probing studies [5–9] and adapter-based designs [10–13]. Yet, control over implicit style remains limited. For example, when using audio to guide symbolic music generation, existing models can extract melody and chords [14, 15], but capturing stylistic traits like the rhythmic feel and more expressive nuances remains a greater challenge. This requires disentangling style from music content, which current music LM-based studies have yet to explore.

In this paper, we explore learning implicit music style in a cross-modal setting for symbolic piano arrangement. Our goal is to generate an expressive piano MIDI performance conditioned on two inputs: an audio example (providing style) and a lead sheet MIDI score (melody and chords as content). To achieve this, we connect pre-trained music LMs in the audio and symbolic domains using a Querying Transformer (Q-Former), a lightweight Transformer originally designed for vision-language alignment [16]. As shown in Figure 1, we extend its role to capture implicit music style, extracting a *style representation* from the hidden states of an audio LM. Then, a symbolic LM conditions on this representation, along with the *content* of the lead sheet, to generate a piano arrangement. The Q-Former

[1] Demo page: https://anonymous55aht.github.io/

© F. Author, S. Author, and T. Author. Licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0). **Attribution:** F. Author, S. Author, and T. Author, "Learning Music Style for Piano Arrangement Through Cross-Modal Bootstrapping", submitted to *ISMIR*, 2026.

enables cross-modal style transfer between two large unimodal LMs without re-training them—a process we refer to as *bootstrapping*.

In our design, we treat the Q-Former as a bottleneck to transfer only style-related information and adopt a two-stage training strategy. The first stage employs contrastive learning, training the Q-Former to extract auditory representations that are musically relevant, expressible in symbolic piano arrangements, and independent of explicit music content. These include accompaniment textures, grooving patterns, and performance dynamics (MIDI velocity contour and tempo). The second stage focuses on generative modeling, where the Q-Former's output conditions the symbolic LM to generate a piano arrangement with the desired expression. We show that the complete system generates more stylistically accurate cover songs compared to existing audio-to-symbolic arrangement methods, while also enabling piano style transfer by conditioning on alternative audio examples. In addition, we demonstrate that the Q-Former can also be applied to audio-to-symbolic retrieval, further highlighting its strength as a general-purpose cross-modal representation learner.

In sum, the contributions of this paper are threefold:

1. We use Q-Former to *align audio and symbolic modalities via implicit music style*, extending its role beyond content alignment in vision-language tasks.

2. We present *a new methodology to disentangle music style from large, pre-trained LMs*, offering a more scalable alternative to traditional latent-variable disentanglement methods.

3. Our model achieves *style-aware audio-to-symbolic piano cover arrangement*. Experiments demonstrate that it outperforms existing audio-to-symbolic models, including both disentanglement-based methods and standard LM approaches.

2. RELATED WORKS

2.1 Music Language Models

Rapid progress in large-scale language models has transformed how we interact with various forms of media, including text, image, and music [16–20]. In particular, large music LMs [4, 21–23] have notably influenced creative practices and user experiences. Models like MusicGen [24] generates music audio with rich timbres directly from text, while MuseCoco [3] produces symbolic compositions with well-structured textures in varied genres. These advancements are driven by training large-scale neural networks on extensive data, scaling up to billions of parameters to enhance controllability and musicality.

Despite these successes, most existing music LMs operate in an unimodal setting, focusing solely on either audio or symbolic representations. Although text-to-music generation has been increasingly effective [4, 21, 22], text descriptions may fall short in expressing nuanced style or performance subtlety. In contrast, our work explores a cross-modal framework that bridges audio and symbolic modalities. By leveraging the strong perceptual understanding of audio LMs and the expressive composition capabilities of symbolic LMs, we bootstrap a system for audio-to-symbolic arrangement. This approach enables more intuitive and fine-grained control over music style beyond what can be conveyed through text alone.

2.2 Piano Cover Generation

Piano cover generation aims to reinterpret an audio recording as a symbolic piano performance. Unlike traditional music transcription, which primarily analyses note-level content such as pitch and timing [25–29], a piano cover often targets higher-level, more structured music elements that shape the *feel* of a performance. The goal is to generate symbolic arrangements that not only sound correct but also feel musically aligned with the original audio.

Existing approaches to piano cover generation often leverage pre-trained transcription models, which primarily extract melodic and harmonic content from the audio [15, 30–33]. However, such models tend to overlook stylistic nuances, resulting in outputs accurate in harmony but lacking the expressive character of the source performance. In this paper, we re-frame piano cover generation through the lens of content-style disentanglement, acquiring *content* in the symbolic form (i.e., melody and chord progression) while learning *style* from the audio. This approach bridges the audio-symbolic gap more effectively, capturing not just *what* is played, but *how* it is played.

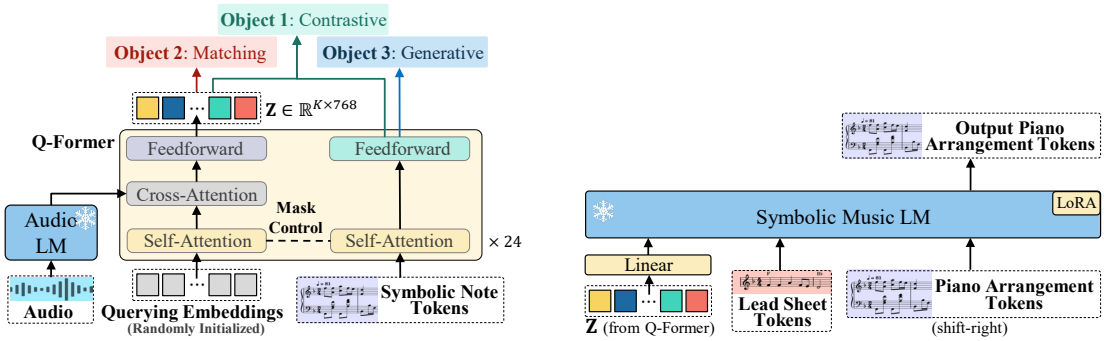
3. METHOD

To bridge the modality gap from audio to symbolic music, we adopt the Q-Former [16] under a two-stage training strategy, as shown in Figure 2. In Section 3.1, we first introduce our audio-symbolic data pairing method that facilitates style learning. We illustrate the Q-Former architecture in Section 3.2, followed by the two-stage training procedure in Sections 3.3 and 3.4. We provide more model configuration details in Appendix A.

3.1 Data Pairing for Style Learning

Training an audio-to-symbolic alignment model requires paired audio–MIDI data. In this work, we use 10-second audio clips paired with 4-bar MIDI segments. Since our goal is to capture style rather than low-level note transcription, we construct the pairs to be *loosely aligned*. Specifically, for each audio clip, we select a MIDI segment near its center with a random temporal shift of up to ± 1 second, and we randomly transpose the MIDI into all 12 keys. This design assumes that music style is locally consistent, while discouraging the model from memorizing exact note-to-note correspondences. In the following sections, we show that the two-step training method enables the model to abstract style features that are shared between the modalities.

We represent music audio as raw waveforms sampled at 32kHz. MIDI is tokenized into note event sequences quantized at 1/12-beat resolution. We include various symbolic



(a) Stage-I: Cross-modal learning with Q-Former.

(b) Stage-II: Audio-to-symbolic arrangement.

Figure 2: Overview of the two-stage framework. (Left) Stage-I: The Q-Former is a Transformer encoder with two parallel, modality-specific streams (audio and symbolic). Both streams are used during training for cross-modal alignment, while only the audio stream is retained at test time. It extracts a cross-modal music style representation $\mathbf{Z}$ via cross-attention to an audio LM. (Right) Stage-II: A symbolic music LM further generates piano arrangements conditioned on (i) the style embedding $\mathbf{Z}$ obtained from the Q-Former and (ii) a lead sheet providing music content.

features including time signature (quadruple and triple meters), tempo curve, note pitch, duration, and velocity.

3.2 Q-Former Architecture

The Q-Former is a Transformer encoder that processes two parallel input streams (audio and symbolic modalities) and learns a shared cross-modal music style representation $\mathbf{Z}$. As shown in Figure 2a, the left stream is connected to the audio LM via cross-attention, while the right stream encodes symbolic piano arrangement tokens via the shared self-attention. A set of K query embeddings (*queries*), initialized randomly, is fed into the left stream to extract style cues from the audio modality while being aligned with the symbolic modality. At test time, only the left stream is retained to uncover cross-modal style directly from audio.

3.3 Stage-I: Audio-Symbolic Representation Learning

We integrate the Q-Former into MusicGen [24], one of the leading audio LMs available today. The queries interact with MusicGen's audio hidden states through cross-attention, and remain connected to the symbolic stream via the shared self-attention layers. To encourage cross-modal style abstraction, we introduce three complementary training objectives, each paired with a tailored self-attention mask that regulates cross-modal interactions, as illustrated in Figure 2a and detailed below.

The primary objective is **Audio-Symbolic Contrastive Learning**, which enforces a higher audio-symbolic similarity for positive (original) pairs compared to negative ones (i.e., randomly paired audio and MIDI clips). Let $\mathbf{Z} \in \mathbb{R}^{K\times 768}$ be the K query outputs from the audio stream of Q-Former, and $t \in \mathbb{R}^{1\times 768}$ be the output embedding of the start token (<s>) from the symbolic stream. We define the audio-symbolic similarity as $\max_k(\cos(\mathbf{Z}_k, t))$ for $k = 1, 2, \cdots, K$, where $\cos(\cdot,\cdot)$ denotes the cosine similarity. The contrastive loss pulls closer aligned audio and symbolic clips in the representation space, while pushing apart unrelated pairs. To prevent information leakage, we employ a *unimodal self-attention mask*, ensuring queries and symbolic tokens do not attend to each other. For detailed mask design, we refer readers to BLIP-2 [16].

The second objective is **Audio-Symbolic Matching**. It is formulated as a binary classification task, where the model predicts whether a given audio-symbolic pair corresponds to each other. On top of contrastive loss, the matching loss aims to capture a finer cross-modal correspondence. In this case, we apply *no masking*, allowing the queries to attend across modalities. Each query output $\mathbf{Z}_k$ is fed into a binary linear classifier to produce a logit, and the logits from all queries are averaged to compute the final matching score. To create informative negative pairs, we employ the hard negative mining strategy in [34].

The final objective, **Audio-Grounded Symbolic Generation**, enforces the Q-Former's right stream to autoregressively reconstruct the input piano arrangement. We implement a *cross-model causal self-attention mask*, allowing the symbolic tokens to attend to the queries but not vice versa. This objective ensures that the style cues extracted from the audio are sufficiently informative to support symbolic realization. To signal a decoding task, we replace the starting <s> token with a special <DEC> token. Additionally, we prepend a sequence of lead sheet tokens before <DEC> so that the queries are encouraged to extract *style*, rather than transcribing *content* from audio.

3.4 Stage-II: Audio-to-Symbolic Generative modeling

In the generative modelling stage, we take advantage of the generative capability of MuseCoco [3], a large-scale symbolic music LM. As illustrated in Figure 2b, MuseCoco is used to reconstruct a piano arrangement based on two concatenated conditional inputs: 1) the query output embeddings $\mathbf{Z}$ from the Q-Former, and 2) a lead sheet. The Q-Former is pre-trained at Stage-I to extract cross-modal music style from the audio, thus providing style guidance. The lead sheet defines the theme melody and chord progression as the content. Since MuseCoco does not natively support lead sheet conditioning, and the inclusion of the lead sheet tokens alters its input format, we insert a LoRA

Table 1: Objective evaluation on *content preservation* and *style coherence*. Ballroom/GTZAN are out-of-distribution sets to assess generalization to unseen genres and instrumentation. Results are reported in the form of mean ± sem^s (in percentage), where sem is the standard error of mean. Different superscript letters s within a column indicate significant differences ($p < 0.05/6$) based on Wilcoxon signed rank test with Bonferroni correction.

	POP909 (In-Distribution) Test Set					Ballroom/GTZAN		
	MCA ↑	**CA ↑**	**GPC ↑**	**VCC ↑**	**TA ↑**	**MCA ↑**	**CA ↑**	**TA ↑**
Ours	32.0±0.5^b	33.3±1.1^b	**79.2**±0.5^a	**76.6**±0.5^a	**83.6**±1.5^a	**17.8**±0.3^a	**16.3**±0.4^a	**79.4**±1.1^a
PCG2	**39.3**±0.6^a	**40.0**±0.9^a	78.4±0.6^a	73.2±0.5^b	74.4±2.0^c	17.2±0.3^a	15.5±0.5^b	57.6±1.5^c
A2M	15.0±0.3^c	22.4±0.9^d	69.0±0.7^b	68.6±0.8^c	77.8±1.4^b	10.9±0.2^b	14.8±0.5^b	-
w/o PT	30.0±0.6^b	28.8±1.0^c	78.6±0.5^a	76.3±0.5^a	68.6±1.9^d	17.2±0.3^a	15.0±0.4^b	74.5±1.3^b

adapter [35] into each self-attention layer. This enables the model to reweight attention and incorporate the new conditioning inputs, while keeping MuseCoco itself frozen.

4. EXPERIMENTS

Our model generates piano performances jointly conditioned on a lead sheet and an audio reference. When the two inputs are aligned with each other, the task corresponds to *piano cover generation*; when they are unpaired, the task becomes *cross-modal style transfer*. This section focus on *piano cover generation*, which allows direct comparison with prior baselines. The other scenario will be covered in Section 5. We introduce the datasets in Section 4.1 and baseline models in Section 4.2. Our evaluation is divided into two parts: objective evaluation in Section 4.3, and subjective evaluation in Section 4.4.

4.1 Datasets

Our model is trained on two dual-modal datasets: POP909 [36] and PIAST [37]. POP909 contains 1K piano cover arrangements created by professional musicians. The music genre is primarily Mandarin pop, while the accompanying audio features diverse band instrumentation, which can help the model learn generalizable audio representations of pop music. PIAST contains 8K piano recordings along with symbolic transcriptions across a variety of genres, including pop, jazz, and classical. Despite the lack of band instrumentation in the piano recording, the genre diversity of PIAST encourages the model to produce more expressive and stylistically varied performances. We split both datasets at song level into training (90%), validation (5%), and test (5%) sets. Each symbolic MIDI file is clipped into 4-bar segments with a 2-bar hop size, center-aligned with the corresponding 10s audio clip.

We also test on two out-of-domain datasets: Ballroom [38, 39] and GTZAN [40, 41], both featuring diverse band/orchestral instrumentation and fine-grained music genres such as jive and bossa nova. Since they lack paired symbolic annotations, we use them for testing only. This allows us to assess the model's generalization ability and its capacity to accommodate styles beyond pop music.

4.2 Baseline Models

We compare our model against two representative piano cover generation models: *PiCoGen2* [32] and *Audio-to-MIDI* [15], as well as one ablation variant of our method.

PiCoGen2 (PCG2) is a Transformer-based language model that builds on the hidden states of Sheetsage [14,42], which itself is derived from Jukebox [43], a pre-trained, large-scale music language model. Leveraging Jukebox's internalized understanding of music *content*, PiCoGen2 generates symbolic piano covers directly from audio.

Audio2MIDI (A2M) is an auto-encoder-based disentanglement framework, using separate modules to extract beat and tempo [44], chord [45], and piano texture from audio. The texture extractor is initialized from a piano transcription model [46]. The extracted components are then merged to form a symbolic piano arrangement.

Ours w/o Pre-Training (w/o PT) is an ablation variant of our model in which the Q-Former is trained directly in Stage-II, without undergoing the representation learning phase in Stage-I. This setup tests the validity of the two-stage training strategy we applied in this work.

To ensure a fair comparison with baseline models, we use Sheetsage [14, 42] to transcribe lead sheets from the audio, making audio the sole input for all methods.

4.3 Objective Evaluation

Piano cover generation transforms an audio input into a symbolic piano performance, which should ideally capture not only *what* is played, but also *how* it is played. In this section, we evaluate *content preservation* (*what* is played) and *style coherence* (*how* it is played) using objective metrics. *Content preservation* assesses whether the melody and harmony are well-maintained. We use *Melody Chroma Accuracy (MCA)* [32, 33] and *Chord Accuracy (CA)* [47, 48] to measure these aspects. *Style coherence* evaluates whether the accompaniment grooves and performance dynamics are well captured and manifested in the symbolic arrangement. We introduce three metrics: *Grooving Pattern Coherence (GPC)* [49], *Velocity Contour Coherence (VCC)*, and *Tempo Accuracy (TA)*. Among them, *GPC* and *VCC* compare the generated covers to human arrangements (available for POP909). *TA* compares the generated tempo to the ground-truth (estimatable from audio using [44] when not annotated). Taken together, these metrics indicate how well the generated cover captures the stylistic "feel" of the reference audio. Across these metrics, a higher value indicates better coherence. Detailed metric definitions are provided in Appendix B.

We consider two evaluation settings: 1) *in-distribution*

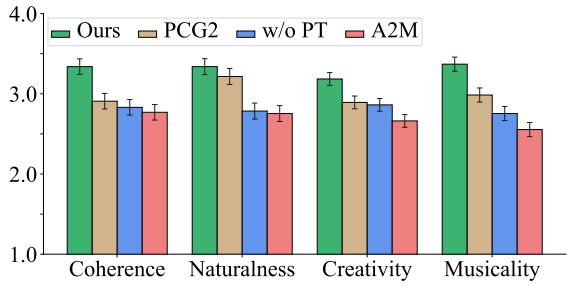


Figure 3: Subjective evaluation on music quality

evaluation on the POP909 test set, and 2) *out-of-distribution* evaluation on 100 tracks randomly drawn from the Ballroom and GTZAN datasets. We run our method and baseline models at each test piece in 10 independent rounds, deriving 450 sets of piano cover samples for POP909, and 1000 sets for Ballroom/GTZAN. We report the mean and standard error. As shown in Table 1, while our method leads in most metrics, we find that on POP909, it achieves a lower *content preservation* score (*MCA* and *CA*) than the *PCG2* baseline. This is probably because our arrangement process relies on Sheetsage-extracted lead sheets, where wrongly estimated chords or melody notes at this stage can propagate to our final results. However, on the out-of-distribution Ballroom/GTZAN dataset, our model surpasses *PCG2* in *MCA* and *CA*, suggesting that *PCG2* is more tailored to pop music (where chords and melodies are relatively simple), whereas our approach generalizes better across diverse music genres and instrumentations. In terms of *style coherence*, our model outperforms all baselines in *GC* and *VC* on POP909, and in *TA* on both test sets, indicating that it can better capture the stylistic characteristics from the audio.

4.4 Subjective Evaluation

We further conduct a double-blind online listening survey to evaluate the music quality. The survey comprises 6 test pieces of varied genres drawn from the Ballroom and the GTZAN datasets. Each test piece is accompanied by 4 piano covers interpreted by our model and each baseline model. For each model, we select the best result from 3 generated samples. The purpose is to prevent occasional low-probability failures (e.g., incomplete or degenerate generations) from disproportionately influencing the subjective assessment. All samples are 16 bars long and rendered to audio using the Cakewalk TTS-1 soundfont, resulting in approximately 40s of audio per sample. Both the order of the test pieces and the order of samples are randomized. Participants are asked to complete 3 test pieces by rating each piano cover on a 5-point Likert scale across 4 criteria: 1) *Audio-to-Symbolic Coherence*, 2) *Naturalness*, 3) *Creativity*, and 4) *Overall Musicality*. More details of the subjective evaluation are provided in Appendix C.

A total of 21 participants with diverse musical backgrounds completed our survey. The average completion time is 12 minutes. Figure 3 shows the mean ratings and standard errors analyzed using within-subject ANOVA [50]. The results reveal significant main effects ($p < 0.05$) across all evaluation criteria. While our model performs comparably to the state-of-the-art *PCG2* in *Naturalness*, it consistently receives higher ratings than the baselines across all criteria. A Bonferroni post-hoc test further confirms that our model significantly outperforms all baselines in *Coherence* and *Musicality*. These results align with the objective evaluation and demonstrate that our model captures music style more effectively and produces coherent, high-quality piano cover arrangements.

Table 2: Objective evaluation on audio-to-symbolic style transfer. The star (*) indicates significant differences ($p < 0.05$) based on the Friedman test.

	MCA	CA	GPC	VCC	TA
Ours	**28.9**±0.5*	**40.0**±1.0*	**64.0**±1.1	**62.8**±1.1	**73.8**±2.0*
w/o PT	25.1±0.5	33.2±1.0	62.9±1.3	62.0±1.3	69.2±2.1

5. ADDITIONAL EVALUATIONS

In this section, we explore additional experimental settings to further evaluate our model's capabilities, with a particular focus on cross-modal representation learning. Specifically, we examine *cross-modal style transfer* in Section 5.1, and audio-to-symbolic retrieval in Section 5.2.

5.1 Evaluation on Cross-Modal Style Transfer

To the best of our knowledge, our method is the first to enable cross-modal audio-to-symbolic style transfer, and thus no established baseline exists for direct comparison. We provided style-varied arrangement demo in Appendix D. In addition, we conduct an objective ablation study comparing our full model (*Ours*) against a variant without Stage-I pretraining (*w/o PT*), to demonstrate the validity of our two-stage training paradigm for cross-modal style transfer.

For this experiment, we collect content lead sheets from the POP909 test split. In 10 independent trials, each lead sheet is paired with a style audio reference randomly drawn from the PIAST test split. This yields 450 style-transfer pairs, and both our full model and the ablation variant generate outputs for each pair. We evaluate *content preservation* using *MCA* and *CA*, which measure melody/chord similarity to the input lead sheet, and *style coherence* using *GPC*, *VCC*, and *TA*, which measure groove/velocity/tempo alignment with the style reference audio's transcription.

As shown in Table 2, our full model outperforms the *w/o PT* variant across all metrics with a good margin, indicating stronger style-transfer capability in both *content preservation* and *style coherence*, thereby validating our two-stage training paradigm. This finding also aligns with the subjective evaluation results in Section 4.4, where our full model is consistently preferred in terms of *Audio-to-Symbolic Coherence*, besides other musicality criteria.

5.2 Evaluation on Audio-to-Symbolic Alignment

While the Q-Former bridges the modality gap for audio-to-symbolic arrangement, it can also operate independently

Table 3: Objective evaluation on audio-to-MIDI retrieval. Results are compared against baseline models and across two evaluation settings: MIDI with random transposition (left) and without transposition (right), highlighting robustness in capturing stylistic features beyond absolute pitch and key.

	w/o Transposition			**w/ Random Transposition**		
	Acc@1 (%) ↑	**Acc@5 (%)** ↑	**Rank** ↓	**Acc@1 (%)** ↑	**Acc@5 (%)** ↑	**Rank** ↓
Random	1.4 ± 0.3	4.8 ± 0.8	64.4 ± 1.7	0.7 ± 0.2	4.2 ± 0.5	65.4 ± 1.0
CLaMP	3.4 ± 0.4	15.0 ± 0.6	42.5 ± 0.2	3.4 ± 0.4	11.0 ± 0.6	48.5 ± 0.3
Ours	$\mathbf{71.4} \pm 1.5$	$\mathbf{95.1} \pm 0.5$	$\mathbf{2.1} \pm 0.1$	$\mathbf{70.2} \pm 1.5$	$\mathbf{94.8} \pm 0.5$	$\mathbf{2.1} \pm 0.1$

as an audio-to-symbolic retriever. In this setting, the Q-Former measures stylistic coherence between an audio clip and a symbolic segment by comparing their learned representations. Given an audio query and a set of symbolic candidates, the model can retrieve the symbolic piece that best aligns with the music style of the query audio. In this section, we compare our approach with CLaMP3 [51] to assess whether our method achieves superior performance.

5.2.1 Audio-to-MIDI Retrieval

To assess the alignment capability of the Q-Former, we evaluate it after Stage-I training on the audio-to-MIDI retrieval task. In each of 10 independent runs, we construct a test set of 128 pairs of 10-second audio and 4-bar MIDI, randomly sampled from PIAST and POP909 (64 pairs each). For each audio query, the model is tasked with retrieving its corresponding MIDI from the full pool of 128 candidates. We consider two evaluation settings: one in which MIDI candidates are randomly transposed to all 12 keys, and the other without transposition. This setup helps us examine the model's robustness in capturing stylistic features beyond absolute pitch and key. Performance is measured using three metrics: *Top-1 Accuracy* (*Acc@1*), *Top-5 Accuracy* (*Acc@5*), and *Mean Rank*. We report the mean and standard error across the 10 resampled runs.

As shown in Table 3, we compare our model against CLaMP3 in addition to a random guessing bot for sanity check. In CLaMP3, audio and MIDI are aligned indirectly via text due to the greater availability of music–text pairs in both domains. While this indirect alignment allows CLaMP3 to perform substantially better than random guessing, its *Acc@1* remains low. Interestingly, transposing the MIDI candidates has a noticeable effect, leading to a 4-point drop in *Acc@5* and a 6-rank increase in *Mean Rank*. This is probably because key signatures are frequently referenced in text descriptions of music, making CLaMP3 particularly sensitive to pitch-level features while struggling to capture finer stylistic nuances. In comparison, our model consistently outperforms CLaMP3 across all metrics and exhibits negligible performance differences between the transposed and non-transposed settings. This demonstrates that our Q-Former learns more robust audio-to-symbolic alignments, effectively capturing stylistic coherence beyond surface-level attributes.

5.2.2 Ablation Study on Pre-Training Objectives

We are also interested in the contribution of each pre-training objective in Section 3.3 to cross-modal alignment. We conduct an ablation study on the Q-Former's audio-to-MIDI retrieval performance based on three different pre-training configurations: contrastive loss only (**C**), contrastive + matching losses (**C+M**), and contrastive + matching + generative losses (**C+M+G**). As in the previous section, we repeat our experiment over 10 independent runs on 128 resampled audio-MIDI pairs.

Table 4: Ablation study on individual pre-training objectives for cross-modal alignment.

	PIAST		**POP909**	
	Acc@1 ↑	**Rank** ↓	**Acc@1** ↑	**Rank** ↓
C	96.7 ± 0.3	1.8 ± 0.2	36.2 ± 1.3	5.2 ± 0.3
C+M	$\mathbf{97.2} \pm 0.4$	$\mathbf{1.6} \pm 0.2$	40.8 ± 0.9	5.0 ± 0.4
C+M+G	97.1 ± 0.3	1.7 ± 0.2	$\mathbf{44.7} \pm 1.4$	$\mathbf{4.7} \pm 0.4$

As shown in Table 4, we conduct evaluation separately on PIAST and POP909. The former involves piano-only music, while the latter includes multi-instrumental accompaniments, requiring the model to extract style from richer audio textures. We observe that the performance difference is relatively small on PIAST, suggesting that contrastive learning alone may suffice for simpler piano alignment. However, on POP909, we see both the matching and generative losses contribute meaningfully to improved retrieval accuracy and lower mean rank. These findings indicate that all three objectives are important for learning robust, generalizable cross-modal alignment.

6. CONCLUSION

In this paper, we introduce a cross-modal framework for audio-to-symbolic arrangement. By repurposing the Q-Former to align audio and symbolic modalities, our model extracts and applies implicit music style using pre-trained music LMs, enabling expressive piano arrangement conditioned on both a lead sheet and an audio reference. Through a two-stage training process—combining representation learning and generative modeling—we extract stylistic features from a frozen, large audio LM and guide a symbolic LM without re-training either backbone. We conduct quantitative experiments on piano cover generation and provide qualitative demos of style transfer. Results demonstrate improved audio-to-symbolic coherence and musicality, highlighting the potential of this framework for controllable, style-aware music generation beyond explicitly labeled content.

7. REFERENCES

[1] R. Yang, D. Wang, Z. Wang, T. Chen, J. Jiang, and G. Xia, "Deep music analogy via latent representation disentanglement," in *Proceedings of the 20th International Society for Music Information Retrieval Conference, ISMIR 2019*, 2019, pp. 596–603.

[2] Z. Wang, D. Wang, Y. Zhang, and G. Xia, "Learning interpretable representation for controllable polyphonic music generation," in *Proceedings of the 21st International Society for Music Information Retrieval Conference, ISMIR 2020*, 2020, pp. 662–669.

[3] P. Lu, X. Xu, C. Kang, B. Yu, C. Xing, X. Tan, and J. Bian, "Musecoco: Generating symbolic music from text," *arXiv preprint arXiv:2306.00110*, 2023.

[4] K. Bhandari, A. Roy, K. Wang, G. Puri, S. Colton, and D. Herremans, "Text2midi: Generating symbolic music from captions," in *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence*. AAAI Press, 2025, pp. 23 478–23 486.

[5] M. Wei, M. Freeman, C. Donahue, and C. Sun, "Do music generation models encode music theory?" in *Proceedings of the 25th International Society for Music Information Retrieval Conference, ISMIR 2024*, 2024, pp. 680–687.

[6] W. Ma and G. Xia, "Exploring the internal mechanisms of music llms: A study of root and quality via probing and intervention techniques," in *ICML 2024 Workshop on Mechanistic Interpretability*, 2024.

[7] W. Ma, X. Li, and G. Xia, "Do music llms learn symbolic concepts? a pilot study using probing and intervention," in *Audio Imagination: NeurIPS 2024 Workshop AI-Driven Speech, Music, and Sound Generation*, 2024.

[8] M. A. V. Vásquez, C. Pouw, J. A. Burgoyne, and W. H. Zuidema, "Exploring the inner mechanisms of large generative music models," in *Proceedings of the 25th International Society for Music Information Retrieval Conference, ISMIR 2024*, 2024, pp. 791–798.

[9] R. Castellon, C. Donahue, and P. Liang, "Codified audio language modeling learns useful representations for music information retrieval," in *Proceedings of the 22nd International Society for Music Information Retrieval Conference, ISMIR 2021*, 2021, pp. 88–96.

[10] L. Lin, G. Xia, J. Jiang, and Y. Zhang, "Content-based controls for music large language modeling," in *Proceedings of the 25th International Society for Music Information Retrieval Conference, ISMIR 2024*, 2024, pp. 783–790.

[11] Y. Zhang, Y. Ikemiya, W. Choi, N. Murata, M. A. Martínez-Ramírez, L. Lin, G. Xia, W.-H. Liao, Y. Mitsufuji, and S. Dixon, "Instruct-musicgen: Unlocking text-to-music editing for music language models via instruction tuning," *arXiv preprint arXiv:2405.18386*, 2024.

[12] L. Lin, G. Xia, Y. Zhang, and J. Jiang, "Arrange, inpaint, and refine: Steerable long-term music audio generation and editing via content-based controls," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI 2024*. ijcai.org, 2024, pp. 7690–7698.

[13] S.-L. Wu, C. Donahue, S. Watanabe, and N. J. Bryan, "Music controlnet: Multiple time-varying controls for music generation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 2692–2703, 2024.

[14] C. Donahue, J. Thickstun, and P. Liang, "Melody transcription via generative pre-training," in *Proceedings of the 23rd International Society for Music Information Retrieval Conference, ISMIR 2022*, 2022, pp. 485–492.

[15] Z. Wang, D. Xu, G. Xia, and Y. Shan, "Audio-to-symbolic arrangement via cross-modal music representation learning," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2022*. IEEE, 2022, pp. 181–185.

[16] J. Li, D. Li, S. Savarese, and S. C. H. Hoi, "BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models," in *International Conference on Machine Learning, ICML 2023*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 19 730–19 742.

[17] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, "Llama: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.

[18] J. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, R. Ring, E. Rutherford, S. Cabi, T. Han, Z. Gong, S. Samangooei, M. Monteiro, J. L. Menick, S. Borgeaud, A. Brock, A. Nematzadeh, S. Sharifzadeh, M. Binkowski, R. Barreira, O. Vinyals, A. Zisserman, and K. Simonyan, "Flamingo: a visual language model for few-shot learning," in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022*, 2022.

[19] Z. Kong, A. Goel, R. Badlani, W. Ping, R. Valle, and B. Catanzaro, "Audio flamingo: A novel audio language model with few-shot learning and dialogue abilities," in *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024.

[20] R. Yuan, H. Lin, S. Guo, G. Zhang, J. Pan, Y. Zang, H. Liu, Y. Liang, W. Ma, X. Du *et al.*, "Yue: Scaling open foundation models for long-form music generation," *arXiv preprint arXiv:2503.08638*, 2025.

[21] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzetti, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi *et al.*, "Musiclm: Generating music from text," *arXiv preprint arXiv:2301.11325*, 2023.

[22] J. Melechovský, Z. Guo, D. Ghosal, N. Majumder, D. Herremans, and S. Poria, "Mustango: Toward controllable text-to-music generation," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), NAACL 2024*. Association for Computational Linguistics, 2024, pp. 8293–8316.

[23] J. Thickstun, D. L. W. Hall, C. Donahue, and P. Liang, "Anticipatory music transformer," *Transactions on Machine Learning Research*, 2024.

[24] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Défossez, "Simple and controllable music generation," in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023*, 2023.

[25] Q. Kong, B. Li, X. Song, Y. Wan, and Y. Wang, "High-resolution piano transcription with pedals by regressing onset and offset times," *IEEE ACM Trans. Audio Speech Lang. Process.*, vol. 29, pp. 3707–3717, 2021.

[26] L. Ou, Z. Guo, E. Benetos, J. Han, and Y. Wang, "Exploring transformer's potential on automatic piano transcription," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2022*. IEEE, 2022, pp. 776–780.

[27] J. Gardner, I. Simon, E. Manilow, C. Hawthorne, and J. H. Engel, "MT3: multi-task multitrack music transcription," in *The Tenth International Conference on Learning Representations, ICLR 2022*. OpenReview.net, 2022.

[28] X. Gu, L. Ou, W. Zeng, J. Zhang, N. Wong, and Y. Wang, "Automatic lyric transcription and automatic music transcription from multimodal singing," *ACM Trans. Multim. Comput. Commun. Appl.*, vol. 20, no. 7, pp. 209:1–209:29, 2024.

[29] W. Zeng, X. He, and Y. Wang, "End-to-end real-world polyphonic piano audio-to-score transcription with hierarchical decoding," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI 2024*. ijcai.org, 2024, pp. 7788–7795.

[30] E. Nakamura and K. Yoshii, "Statistical piano reduction controlling performance difficulty," *APSIPA Transactions on Signal and Information Processing*, vol. 7, 2018.

[31] C. Tan, S. Guan, and Y. Yang, "Picogen: Generate piano covers with a two-stage approach," in *Proceedings of the 2024 International Conference on Multimedia Retrieval, ICMR 2024*. ACM, 2024, pp. 1180–1184.

[32] C. Tan, H. Ai, Y. Chang, S. Guan, and Y. Yang, "Picogen2: Piano cover generation with transfer learning approach and weakly aligned data," in *Proceedings of the 25th International Society for Music Information Retrieval Conference, ISMIR 2024*, 2024, pp. 555–562.

[33] J. Choi and K. Lee, "Pop2piano : Pop audio-based piano cover generation," in *IEEE International Conference on Acoustics, Speech and Signal Processing ICASSP 2023*. IEEE, 2023, pp. 1–5.

[34] J. Li, R. R. Selvaraju, A. Gotmare, S. R. Joty, C. Xiong, and S. C. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021*, 2021, pp. 9694–9705.

[35] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," in *The Tenth International Conference on Learning Representations, ICLR 2022*. OpenReview.net, 2022.

[36] Z. Wang, K. Chen, J. Jiang, Y. Zhang, M. Xu, S. Dai, and G. Xia, "POP909: A pop-song dataset for music arrangement generation," in *Proceedings of the 21st International Society for Music Information Retrieval Conference, ISMIR 2020*, 2020, pp. 38–45.

[37] H. Bang, E. Choi, M. Finch, S. Doh, S. Lee, G.-H. Lee, and J. Nam, "Piast: A multimodal piano dataset with audio, symbolic and text," in *Proceedings of the 3rd Workshop on NLP for Music and Audio (NLP4MusA)*, 2024, pp. 5–10.

[38] F. Gouyon, A. Klapuri, S. Dixon, M. Alonso, G. Tzanetakis, C. Uhle, and P. Cano, "An experimental comparison of audio tempo induction algorithms," *IEEE Trans. Speech Audio Process.*, vol. 14, no. 5, pp. 1832–1844, 2006.

[39] F. Krebs, S. Böck, and G. Widmer, "Rhythmic pattern modeling for beat and downbeat tracking in musical audio," in *Proceedings of the 14th International Society for Music Information Retrieval Conference, ISMIR 2013*, 2013, pp. 227–232.

[40] G. Tzanetakis and P. R. Cook, "Musical genre classification of audio signals," *IEEE Trans. Speech Audio Process.*, vol. 10, no. 5, pp. 293–302, 2002.

[41] U. Marchand and G. Peeters, "Swing ratio estimation," in *Proceedings of the 18th International Conference on Digital Audio Effects, DAFx-15*, 2015, pp. 1–6.

[42] C. Donahue and P. Liang, "Sheet sage: Lead sheets from music audio," *ISMIR 2021 Late-Breaking and Demo*, 2021.

[43] P. Dhariwal, H. Jun, C. Payne, J. W. Kim, A. Radford, and I. Sutskever, "Jukebox: A generative model for music," *arXiv preprint arXiv:2005.00341*, 2020.

[44] S. Böck and M. E. P. Davies, "Deconstruct, analyse, reconstruct: How to improve tempo, beat, and downbeat estimation," in *Proceedings of the 21th International Society for Music Information Retrieval Conference, ISMIR 2020*, 2020, pp. 574–582.

[45] J. Jiang, K. Chen, W. Li, and G. Xia, "Large-vocabulary chord transcription via chord structure decomposition," in *Proceedings of the 20th International Society for Music Information Retrieval Conference*, 2019, pp. 644–651.

[46] C. Hawthorne, E. Elsen, J. Song, A. Roberts, I. Simon, C. Raffel, J. H. Engel, S. Oore, and D. Eck, "Onsets and frames: Dual-objective piano transcription," in *Proceedings of the 19th International Society for Music Information Retrieval Conference, ISMIR 2018*, 2018, pp. 50–57.

[47] Y. Ren, J. He, X. Tan, T. Qin, Z. Zhao, and T.-Y. Liu, "Popmag: Pop music accompaniment generation," in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 1198–1206.

[48] J. Zhao, G. Xia, Z. Wang, and Y. Wang, "Structured multi-track accompaniment arrangement via style prior modelling," in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024*, 2024.

[49] S. Wu and Y. Yang, "The jazz transformer on the front line: Exploring the shortcomings of ai-composed music through quantitative measures," in *Proceedings of the 21st International Society for Music Information Retrieval Conference*, 2020, pp. 142–149.

[50] H. Scheffe, *The analysis of variance*. John Wiley & Sons, 1999, vol. 72.

[51] S. Wu, Z. Guo, R. Yuan, J. Jiang, S. Doh, G. Xia, J. Nam, X. Li, F. Yu, and M. Sun, "Clamp 3: Universal music information retrieval across unaligned modalities and unseen languages," in *Findings of the Association for Computational Linguistics, ACL 2025*. Association for Computational Linguistics, 2025, pp. 2605–2625.

[52] M. Zeng, X. Tan, R. Wang, Z. Ju, T. Qin, and T. Liu, "Musicbert: Symbolic music understanding with large-scale pre-training," in *Findings of the Association for Computational Linguistics: ACL/IJCNLP 2021*, ser. Findings of ACL, vol. ACL/IJCNLP 2021. Association for Computational Linguistics, 2021, pp. 791–800.

[53] Y.-S. Huang and Y.-H. Yang, "Pop music transformer: Beat-based modeling and generation of expressive pop piano compositions," in *MM '20: The 28th ACM International Conference on Multimedia*, 2020, pp. 1180–1188.

[54] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *7th International Conference on Learning Representations, ICLR 2019*. OpenReview.net, 2019.

[55] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, and D. P. W. Ellis, "Mir_eval: A transparent implementation of common MIR metrics," in *Proceedings of the 15th International Society for Music Information Retrieval Conference, ISMIR 2014*, 2014, pp. 367–372.

[56] S. Rouard, F. Massa, and A. Défossez, "Hybrid transformers for music source separation," in *IEEE International Conference on Acoustics, Speech and Signal Processing ICASSP 2023*. IEEE, 2023, pp. 1–5.

[57] M. Mauch and S. Dixon, "PYIN: A fundamental frequency estimator using probabilistic threshold distributions," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2014*. IEEE, 2014, pp. 659–663.

[58] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," in *Proceedings of the 14th Python in Science Conference 2015 (SciPy 2015)*. scipy.org, 2015, pp. 18–24.

[59] A. L. Uitdenbogerd and J. Zobel, "Manipulation of music for melody matching," in *Proceedings of the 6th ACM International Conference on Multimedia '98*. ACM, 1998, pp. 235–240.

[60] J. Jiang, "MIDI Chord Recognition via Bar-Level Modeling," https://github.com/music-x-lab/midi-chord-recognition, 2025.

[61] S. Dixon, "Automatic extraction of tempo and beat from expressive performances," *Journal of New Music Research*, vol. 30, no. 1, pp. 39–58, 2001.

A. MODEL AND TRAINING DETAILS

Our model comprises three components: an audio LM, a symbolic music LM, and a Q-Former connecting the two. This section provides detailed configurations of each component module and the training details.

A.1 Q-Former

The Q-Former is intialized with the MusicBERT-Base model [52]. The added cross-attention layers are randomly initialized. Following Blip-2 [16], we use $K = 32$ learnable queries, each with dimension 768. Symbolic piano arrangements are tokenized in the OctMIDI format [52], which produces note-wise joint embeddings. Overall, the Q-Former comprises 186M parameters, including the learnable queries and note embedding layers.

A.2 Audio LM

We use MusicGen-Large [24] as our audio LM. We discard the text encoder and retain only the music decoder, a 48-layer Transformer. Audio codecs are fed to the decoder and we extract the hidden representations from the 25th layer, as prior probing studies [5, 7–9] suggest that middle layers capture more musically meaningful features. This setup retains 1.7B frozen parameters from MusicGen.

A.3 Symbolic Music LM

For symbolic music arrangement, we adopt MuseCoco-xLarge [3], which is a 24-layer Transformer decoder pre-trained on large-scale symbolic music corpora. We remove its text-related components and keep 1.2B frozen parameters from the music decoder. Symbolic note tokens are converted into the REMI format [53]. Despite the slightly different tokenizations used across stages, we find that the latent representations learned in Stage-I remain compatible with Stage-II. To enable compatibility with MuseCoco, we project $\mathbf{Z}$ into the same embedding dimension as MuseCoco's token embeddings via a linear layer. A LoRa adapter with rank 16 accomodates the added lead sheet condition.

A.4 Training Details

Model training focuses on the 186M Q-Former parameters, which is significantly smaller than the billion-scale LM backbones. In Stage-I, the Q-Former is pre-trained in FP16 using batch size 128 for 10 epochs (130K iteration). The LoRA adaptor in Stage-II adds 5M parameters and we fine-tune the model for another 5 epochs using batch size 32. Both training stages are conducted on four RTX A40 GPUs (48GB each). We use the AdamW optimizer [54] with an initial learning rate of 1e-4, a linear warm-up over the first 1k steps, and a cosine decay schedule to a final rate of 1e-5. At test time, we use top-k sampling with $k = 15$.

B. OBJECTIVE METRICS

We introduce five statistical metrics to evaluate *content preservation* and *style coherence* for the piano cover generation tasks. This section provides the definitions.

B.1 Melody Chroma Accuracy (MCA)

We use the `melody.raw_chroma_accuracy` metric [2] provided by mir_eval [55] to evaluate the similarity between two monophonic melody sequences. For the reference melody, we apply Demucs [56] to isolate the vocal stem from the audio and then extract the F0 contour using pYIN [57] provided by librosa [58]. For the estimated melody, we obtain the melody skyline [59] from the generated piano cover MIDI and convert the MIDI pitches to frequencies. The two melodies are compared position-wise under a tolerance of 50 cents. While this implementation follows [32], we additionally ensure that the two melodies are temporally aligned to the same sequence length.

B.2 Chord Accuracy (CA)

We introduce *Chord Accuracy* from [47, 48] to measure the similarity between two chord sequences. For the reference sequence, when annotated chords are not available, we apply the method of [45] to detect chords from the input audio. For the estimated sequence, we use [60] to detect chords from the generated piano cover MIDI. Both chord sequences are aligned and compared at 1-beat granularity in terms of root and full quality based on the MIREX TeTrads rule. [3]

B.3 Groove Pattern Coherence (GPC)

Grooving Pattern Coherence (*GPC*) evaluates the *grooving pattern* similarity between the generated piano cover and human's arrangement. The *grooving pattern*, which is defined in [49], represents the positions in a MIDI segment at which there is at least one note onset. We consider 4-bar segments at 1/4-beat granularity, deriving *grooving pattern* $\mathbf{g}$ as a 64-dimensional binary vector. The *GPC* over a test piece is defined as follows:

$$GPC = \frac{1}{N}\sum_{n=1}^{N}\cos(\mathbf{g}_n^{\mathrm{hm}}, \mathbf{g}_n^{\mathrm{cv}}), \tag{1}$$

where N is the number of non-overlapping 4-bar segments in the test piece. $\cos(\cdot,\cdot)$ computes the cosine similarity. $\mathbf{g}^{\mathrm{hm}}$ and $\mathbf{g}^{\mathrm{cv}}$ represent the *grooving pattern* feature from the human arrangement and the generated piano cover, respectively. The *GPC* metric is omitted for evaluation on Ballroom/GTZAN, where human's piano arrangement is not available.

B.4 Velocity Contour Coherence (VCC)

Velocity Contour Coherence (*VCC*) evaluates the similarity of the *velocity contour* between the generated piano cover and the human arrangement, using the same formulation as *GPC*. We define the *velocity contour* as a time-series feature representing the average note velocity at each

[2] https://mir-eval.readthedocs.io/latest/api/melody.html #mir_eval.melody.raw_chroma_accuracy

[3] https://mir-eval.readthedocs.io/latest/api/chord.html #mir_eval.chord.tetrads

timestep. It has the same dimensionality and temporal granularity as the *grooving pattern*, but instead of a binary vector, it consists of real values ranging from 0 to 128. The *VCC* metric is omitted for Ballroom/GTZAN, where human's piano arrangement is not available.

B.5 Tempo Accuracy (TA)

Tempo Accuracy (*TA*) evaluates the correctness of the estimated tempo $\mathrm{tp}^{\mathrm{est}}$ relative to the reference (ground-truth) tempo $\mathrm{tp}^{\mathrm{ref}}$. We first define correctness indicator $D(k)$ for one test piece as follows:

$$D(k) = \mathbb{1}\left(\frac{|\mathrm{tp}^{\mathrm{ref}} - k \times \mathrm{tp}^{\mathrm{est}}|}{\mathrm{tp}^{\mathrm{ref}}} < 0.08\right), \qquad (2)$$

where $\mathbb{1}(\cdots)$ is the indicator function. $k \in \{1/2, 1, 2\}$ takes into account the octave ambiguity. The tolerance threshold of 0.08 follows the empirical setting in mir_eval. [4]

We then define tempo accuracy *TA* as:

$$\mathit{TA} = \max\Big(D(1),\ 0.5 \cdot D(1/2),\ 0.5 \cdot D(2)\Big), \qquad (3)$$

where we consider half-tempo and double-tempo matches as partially correct (weight 0.5). This is because tempo perception is known to exhibit octave ambiguity, where perceptions at multiple metrical levels are still considered valid rather than true perceptual errors [61].

We derive the ground-truth tempo from beat annotations when available. On Ballroom/GTZAN, we use the audio tempo estimation results by madmom [44], and we omit the *A2M* baseline in this setting because it already relies on madmom's estimation in its pipeline.

C. SUBJECTIVE EVALUATION DETAILS

Our subjective evaluation is conducted through an online crowdsourcing study, where participants complete a survey consisting of listening and rating tasks. This section provides additional details on the survey design.

C.1 General Instructions

Participants receive the general instructions, which clarify their rights and the conditions of participation:

- Participation is entirely voluntary, and one may withdraw at any time without any negative consequences.
- No personally identifying information is collected; all responses are anonymous and used solely for research purposes.

[4] https://mir-eval.readthedocs.io/latest/api/tempo.html #mir_eval.tempo.detection

C.2 Survey Design

Our survey consists of 6 pages, each presenting 4 versions of piano cover arrangements corresponding to a common test audio piece. The test audio pieces are drawn from the Ballroom and GTZAN datasets and span a variety of genres. The 4 arrangement versions are produced by our model and all baseline models (*PCG2*, *A2M*, and *w/o PT*), respectively. For each model, we select the best result from 3 independently generated samples to avoid occasional low-probability failures (e.g., incomplete or degenerate outputs) from disproportionately affecting subjective evaluation. All models to be evaluated are anonymized, and their orders on each page are randomized.

We acknowledge that participants may use different internal criteria when evaluating music. To minimize such variability, we conduct within-subject ANOVA [50] for data analysis, which ensures that variances in ratings reflect differences among the models rather than participants. To control response quality, we only accept complete evaluation sets; that is, participants have to rate all four models under a common test piece for their responses to be considered valid.

C.3 Completion Time

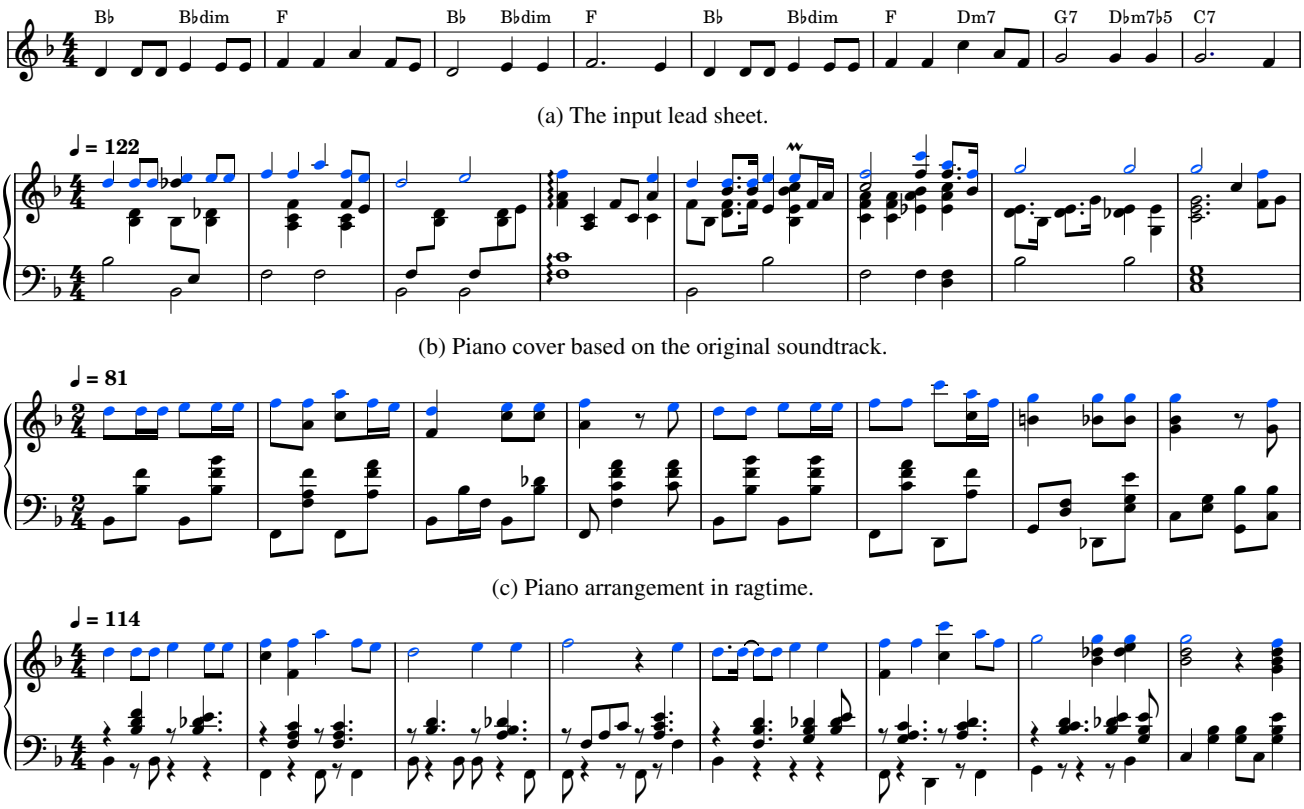
Each participant is randomly assigned 3 out of the 6 pages. On each page, participants first listen to the test audio piece, and then listen to and rate 4 corresponding piano cover samples. All samples in the survey are 16 bars long and rendered to audio using the Cakewalk TTS-1 soundfont, producing approximately 40 seconds of audio per sample. This design targets a total completion time of 10–15 minutes, ensuring that participants have sufficient time to listen carefully without excessive fatigue. The actual completion time observed on average is 12 minutes.

C.4 Participant Profiles

Participants are asked to self-identify their musical background as *amateur*, *intermediate*, or *professional*, following the guidelines below:

- **Amateur**: I enjoy listening to music. I can play/sing/compose short music pieces. I know a little music theory. I can evaluate a composition based on my feelings.
- **Intermediate**: I have some experience in performing, composing, or other music activities. I know a certain amount of music theories that can help me evaluate a composition.
- **Professional**: I am now pursuing/have completed a music degree, or having equivalent background. I am proficient in using music theory to evaluate a composition.

Among the 21 valid responses collected, 6 participants identified as *amateur* (28.6%), 11 as *intermediate* (52.4%), and 4 as *professional* (19.0%). All authors are excluded from the survey.



(a) The input lead sheet.

(b) Piano cover based on the original soundtrack.

(c) Piano arrangement in ragtime.

(d) Piano arrangement in bossa nova.

Figure 4: Audio-to-symbolic arrangement for an 8-bar excerpt from *The Sound of Music*. Score is manually engraved from MIDI for visualisation purpose. Figures 4b to 4d are arranged based on the lead sheet in 4a and an audio reference from the original soundtrack, a ragtime piece, and a bossa nova piece, respectively. Preserved music contents are highlighted in blue note heads. Synthesized audio is provided on the demo page: `https://anonymous55aht.github.io/`.

D. ARRANGEMENT DEMONSTRATION

In this section, we demonstrate the performance of our audio-to-symbolic arrangement model under *freely manipulated* audio style references. Figure 4a shows an 8-bar lead sheet excerpt from the musical *The Sound of Music*. The selected passage features harmonically rich chords, including diminished and seventh chord qualities, which present suitable complexity for arrangement experiments. Figures 4b to 4d showcase the arrangement results conditioned on varied audio references. The 8-bar arrangement is generated using windowed sampling, wherein a 4-bar context window progresses forward every 2 bars and continues sampling conditioned upon the preceding 2 bars.

Figure 4b shows the piano cover from the original *The Sound of Music* soundtrack,[5] which features lush orchestration dominated by string ensembles. Our arrangement captures this orchestral essence through dense, block-chord voicing that emulates the sonority of string sections. Additionally, ornaments such as arpeggios and trills are found to complement the sweeping harmonic textures, which contributes to the music's free-flowing character.

Figure 4c shows an arrangement conditioned on the ragtime classic *The Entertainer*.[6] Following the audio recording, the arrangement's tempo is "not fast," and the piano texture distinctly adopts a ragtime rhythm, with steady bass notes on downbeats and syncopated chordal accents on upbeats.

Figure 4d shows an arrangement conditioned on the bossa nova piece *The Girl from Ipanema*.[7] In this interpretation, the arrangement is characterized by a moderate tempo and distinctive left-hand syncopated patterns characteristic of the bossa nova genre.

Across all three piano arrangements, while distinct music styles are effectively captured from the audio references, the theme melody and harmonic structures remain faithfully preserved. In Figure 4, we highlight melody notes preserved from the lead sheet using blue note heads.

Additional examples are available on our demo page,[8] including arrangements in a wider range of classical, jazz, and pop styles applied to well-known lead sheets, illustrating the flexibility of both style and lead sheet control.

E. LIMITATION

Our proposed method demonstrates the ability to learn implicit music style from audio. At the current stage of this work, we acknowledge that the extracted style

[5] Original audio: `https://youtu.be/6f0T6UV-HiI&t=57`
[6] Ragtime audio: `https://youtu.be/jKlfNfRZL9I&t=11`
[7] Bossa nova audio: `https://youtu.be/DvA_wDOVD10&t=12`
[8] Demo page: `https://anonymous55aht.github.io/`

primarily represents *segment-level global* characteristics. Here, "global" refers to a latent style profile at the level of short 4-bar segments, which is still considerably more fine-grained (and more local) than typical global attributes such as genre labels and textual style tags. Our underlying assumption is that music style remains locally consistent at the segment or bar level. We empirically found that this assumption holds across most music traditions, which guarantees our model to perform reliably under this design. While longer generation can be achieved using windowed sampling, we acknowledge that this approach may smooth over intended stylistic transitions at phrase boundaries, thus leading to diminished expressivity at longer timescales. When considering inter-phrase and longer-term music development, we recognize that a single segment-level style representation is insufficient to capture evolving dynamics. Also, subject to the availability of audio-symbolic data, this work is dedicated to piano arrangement. The cross-modal arrangement of long-term, multi-track music may require hierarchical or temporally adaptive style modelling, which is an important direction for our future work.